Introduction to Javascript

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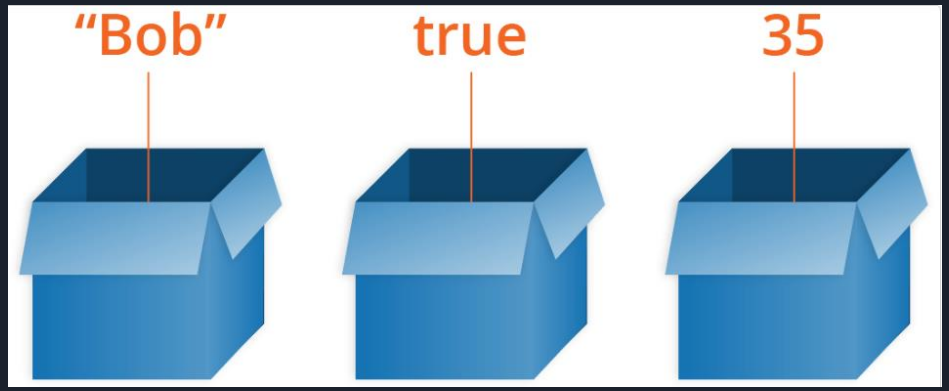
Basics of Javascript

JavaScript is a powerful programming language that can add interactivity to a website. It was invented by Brendan Eich.

JavaScript is a programming language that adds interactivity to your website. This happens in games, in the behavior of responses when buttons are pressed or with data entry on forms; with dynamic styling; with animation, etc.

Variables

A variable is a container for a value, like a number we might use in a sum, or a string that we might use as part of a sentence.



Declaring a Variable

JavaScript Variables can be declared in 4 ways:

- Using var
- Using let
- Using const

Var a = 5

Let b = 6

Const c = 8

Let and Const

Variables defined with let cannot be Redeclared

Variables defined with let must be Declared before use

Variables defined with let have Block Scope

Variables defined with const cannot be Redeclared

Variables defined with const cannot be Reassigned

Variables defined with const have Block Scope

When to use var,let and const

Always declare variables

2. Always use const if the value should not be changed

3. Always use const if the type should not be changed (Arrays and Objects)

4. Only use let if you can't use const

5. Only use var if you MUST support old browsers.

Rules for naming a variable

The general rules for constructing names for variables (unique identifiers) are:

- Names can contain letters, digits, underscores, and dollar signs.
- Names must begin with a letter.
- Names can also begin with $ and _ (but we will not use it in this tutorial).
- Names are case sensitive (y and Y are different variables).
- Reserved words (like JavaScript keywords) cannot be used as names.

Data Types

JavaScript has 8 Datatypes

1. String
2. Number
3. Bigint
4. Boolean
5. Undefined
6. Null
7. Symbol
8. Object

Operators

There are different types of JavaScript operators:

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- String Operators
- Logical Operators
- Bitwise Operators
- Ternary Operators

Arithmetic Operators

Operator	Description
+	Addition
-	Subtraction
*	Multiplication
**	Exponentiation (ES2016)
/	Division
%	Modulus (Division Remainder)
++	Increment
--	Decrement

Assignment Operators

Operator	Example	Same As
=	x = y	x = y
+=	x += y	x = x + y
-=	x -= y	x = x - y
*=	x *= y	x = x * y
/=	x /= y	x = x / y
%=	x %= y	x = x % y
**=	x **= y	x = x ** y

Comparison Operators

Operator	Description
==	equal to
===	equal value and equal type
!=	not equal
!==	not equal value or not equal type
>	greater than
<	less than
>=	greater than or equal to
<=	less than or equal to
?	ternary operator

Logical Operators

Operator	Description
&&	logical and
||	logical or
!	logical not

Loops

Loops can execute a block of code a number of times.

JavaScript supports different kinds of loops:

- for - loops through a block of code a number of times
- for/in - loops through the properties of an object
- for/of - loops through the values of an iterable object
- while - loops through a block of code while a specified condition is true
- do/while - also loops through a block of code while a specified condition is true

For loop

The for statement creates a loop with 3 optional expressions:

```
for (let i = 0; i < 5; i++) {
  text += "The number is " + i + "<br>";
}
```

While loop

The while loop loops through a block of code as long as a specified condition is true.

```
while (i < 10) {
  text += "The number is " + i;
  i++;
}
```

For in loop

The JavaScript for in statement loops through the properties of an Object

```
const person = {fname:"John", lname:"Doe", age:25};


let text = "";
for (let x in person) {
  text += person[x];
}
```

Q & A